

Heat, Smoke & CO Detectors: Compliance, SOP & Optimization Guide

Purpose:

Detects temperature rise, smoke particles, or carbon monoxide to trigger early warning systems—essential for preventing injuries

Why It's Ignored:

Devices are small, quiet, and often assumed to be functional until they fail. Batteries may be removed or never replaced.

Must-Have Docs:

- Detector zone map and type registry
- Functional test records (monthly/quarterly)
- Replacement/battery change logs
- Calibration certificates (for CO detectors)
- Evacuation drill reports

Compliance Officials:

NFPA 72, OSHA 1910.165, Local Fire Marshal or Building Inspector

PPE Required:

None under normal conditions. When testing or replacing: fall protection (for lifts), gloves, possibly arc-rated PPE if near electrical panels

Installation Practices:

- Smoke: mounted on ceiling ≥ 4 " from wall, away from air vents
- Heat: above fire risk zones (boilers, kitchens)
- CO: ~5 ft above floor in sleeping or occupied areas
- Hardwired + battery backup for new installs
- Clearly labeled, documented, and tagged with test dates

Must Checks:

- Monthly test button activation (log it)
- Battery voltage or replacement confirmation
- Detector cleaning (to remove dust or insect nests)
- Response time during drills
- CO ppm levels during simulated or real events

Regulations & Codes:

NFPA 72, OSHA 1910.165, IBC and IFB, Manufacturer specs for CO calibration